FACULTY OF COMPUTING

SEMESTER 1

2024/2025

SECR1213-02 NETWORK COMMUNICATIONS

LECTURER: DR. ISMAIL FAUZI BIN ISNIN

GROUP NAME: Y-LINK

NAME	MATRIC NUMBER
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1.0 Get the Network Address from your Lecturer

Figure 1.1 shows the list of network IP addresses assigned to each group. The network address of our group is 10.64.0.0/12.

Network IP address

Group 1 = 10.16.0.0/12
Group 2 = 10.32.0.0/12
Group 3 = 10.48.0.0/12
Group 4 = 10.64.0.0/12
Group 5 = 10.80.0.0/12
Group 6 = 10.96.0.0/12
Group 7 = 10.112.0.0/12
Group 8 = 10.128.0.0/12
Group 9 = 10.144.0.0/12
Group 10 = 10.176.0.0/12

Figure 1.1

Table 1.0 shows the network details we gain from a given IP address. From the IP address, we determine the IP address in binary, subnet mask in binary and subnet mask in decimal.

Table 1.1 : IP Address Details

IP address	10.64.0.0/12
IP address(binary)	00001010.01000000.00000000.00000000
Subnet Mask(binary)	11111111.11110000.00000000.00000000
Subnet Mask(decimal)	255.240.0.0

We determine the network address and broadcast IP address by performing AND operation of the IP address with subnet mask as seen in Figure. The network address we found is 10.64.0.0 and 10.64.255.255 for broadcast IP address. The range of usable IP addresses is between 10.64.0.1 to 10.64.255.254. Table 1.2 shows the details of the result of the AND operation.

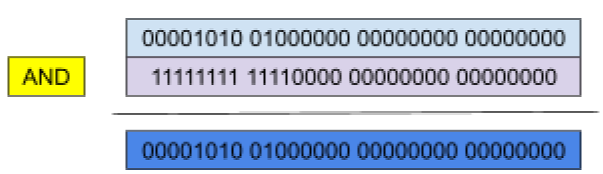

Figure 1.2 : AND operation

Table 1.2 : Result of AND operation

Network address	10.64.0.0
Broadcast IP address	10.64.255.255
Range of Usable IP addresses	10.64.0.1 - 10.64.255.254

2.0 Divide it in the Best Possible Way for your Network

Table 1.3 shows the host needed for each work area to help in dividing the network through each room and labs.

Table 1.3 : Host needed for Work Area

Work area	Host needed
1	5
2	2
3	24
4	2
5	12
6	14
7	31
8	31
9	32
10	33
11	42

We decided to have 11 subnets since the building has 11 work areas. Subnet building requires borrowing 4 bits (2^4 = 16) from the host portion in order to potentially achieve 11 subnets. Thus, /16 is our new subnet mask. Table 1.4 and Table 1.5 shows the distribution of the network to each work area and IP address assigned to each room and lab respectively.

Table 1.4 : Distribution of Network to each Work Area

Subnet	Network Portion	Subnet bits	Host Portion		Custom Subnet Mask (slash format and dotted decimal format)
0	00001010 01000000	0000	0000 00000000	Network address	10.64.0.0
			1111 11111111	Broadcast address	10.64.15.255
1	00001010 01000000	0001	0000 00000000	Network address	10.64.16.0
			1111 11111111	Broadcast address	10.64.31.255
2	00001010 01000000	0010	0000 00000000	Network address	10.64.32.0
			1111 11111111	Broadcast address	10.64.47.255
3	00001010 01000000	0011	0000 00000000	Network address	10.64.48.0
			1111 11111111	Broadcast address	10.64.63.255
4	00001010 01000000	0100	0000 00000000	Network address	10.64.64.0
			1111 11111111	Broadcast address	10.64.79.255
5	00001010 01000000	0101	0000 00000000	Network address	10.64.80.0
			1111 11111111	Broadcast address	10.64.95.255
6	00001010 01000000	0110	0000 00000000	Network address	10.64.96.0

			1111 11111111	Broadcast address	10.64.111.255
7	00001010 01000000	0111	0000 00000000	Network address	10.64.112.0
			1111 11111111	Broadcast address	10.64.127.255
8	00001010 01000000	1000	0000 00000000	Network address	10.64.128.0
			1111 11111111	Broadcast address	10.64.143.255
9	00001010 01000000	1001	0000 00000000	Network address	10.64.144.0
			1111 11111111	Broadcast address	10.64.159.255
10	00001010 01000000	1010	0000 00000000	Network address	10.64.160.0
			1111 11111111	Broadcast address	10.64.175.255

Table 1.5 : IP Address Assigned to Each Room and Lab

Subnet	Work Area	Network Address	Broadcast Address	Range of Usable Addresses	Custom Subnet Mask (slash format and dotted decimal format)
0	Video Conference Room	10.64.0.0	10.64.15.255	10.64.0.1 - 10.64.15.254	/16 255.255.0.0
1	Student Lounge	10.64.16.0	10.64.31.255	10.64.16.1 - 10.64.31.254	/16 255.255.0.0
2	Hybrid Classroom	10.64.32.0	10.64.47.255	10.64.32.1 - 10.64.47.254	/16 255.255.0.0
3	Prayer Room	10.64.48.0	10.64.63.255	10.64.48.1 - 10.64.63.254	/16 255.255.0.0
4	Server Room	10.64.64.0	10.64.79.255	10.64.64.1 - 10.64.79.254	/16 255.255.0.0
5	Staff Room (Ground Floor)	10.64.80.0	10.64.95.255	10.64.80.1 - 10.64.95.254	/16 255.255.0.0
6	General Purpose Lab 1	10.64.96.0	10.64.111.255	10.64.96.1 - 10.64.111.254	/16 255.255.0.0
7	General Purpose Lab 2	10.64.112.0	10.64.127.255	10.64.112.1 - 10.64.127.254	/16 255.255.0.0
8	Embedded Lab	10.64.128.0	10.64.143.255	10.64.128.1 - 10.64.143.254	/16 255.255.0.0
9	Cisco Network Lab	10.64.144.0	10.64.159.255	10.64.144.1 - 10.64.159.254	/16 255.255.0.0
10	Staff Room (First Floor)	10.64.160.0	10.64.175.255	10.64.160.1 - 10.64.175.254	/16 255.255.0.0

MINUTES MEETING

Date/Time	22 Dec 2024, 10.25 p.m.	
Location	Online (Google Meet)	
Agenda	1. Discuss the questions on Task 5	
	2. Task distribution for each person	
Meeting Host	Safiya	
Attendance		
Name	Time	Reason For Absence
Dayang Farah Farzana	10.25 p.m.	-
Farra Nurzahin	10.25 p.m.	-
Harini	10.25 p.m.	-
Safiya Nursyahadah	10.25 p.m.	-

Minutes			
No.	Item Discussed	Details	Person in charge
1.	A short discussion on question given on Task 5	- All members read the question and rubric together.	All members
2.	Identify the numbers of subnet	- All members count the number of subnets needed for each floor.	All members
3.	Divide network to all different labs and room	- All members discuss how to divide the network and assign IP addresses.	All members
4.	Meeting ended	- At 11.30 p.m., Dayang ends the meeting after all discussions have been done.	Dayang